In[30]:=
```
Clear["Global`*"];
```

In[31]:=
```
Clear[K];
"Standard Normal CDF";
Φ[x_] = CDF[NormalDistribution[0, 1], x];

"Black-Scholes Call price function in
  terms of log-moneyness and v=σ^2 T with S0=1,r=0";
BSCall[x_, v_] = Φ[(-x + 1/2 v) / Sqrt[v]] - Exp[x] Φ[(-x - 1/2 v) / Sqrt[v]];

"Assume F0=1";
"SVI Call price function";
SVICall[K_] = BSCall[ Log[K], V[Log[K]]];
"SVI Density using Breeden-Litzenberger";
q[K_] = Simplify[D[SVICall[K], K, K]]
```

Out[28]=

$$-\left(\left(e^{-\frac{4\,\mathrm{Log}[K]^2+V[\mathrm{Log}[K]]^2}{8\,V[\mathrm{Log}[K]]}}\left(-4\,\mathrm{Log}[K]^2\,V'[\mathrm{Log}[K]]^2+4\,V[\mathrm{Log}[K]]\,V'[\mathrm{Log}[K]]\,\left(4\,\mathrm{Log}[K]+V'[\mathrm{Log}[K]]\right)+V[\mathrm{Log}[K]]^2\,\left(V'[\mathrm{Log}[K]]^2-8\,\left(2+V''[\mathrm{Log}[K]]\right)\right)\right)\right)\Big/\left(16\,K^{3/2}\,\sqrt{2\,\pi}\,V[\mathrm{Log}[K]]^{5/2}\right)\right)$$

In[•]:=
```
"SVI formula";
V[x_] = a + b (ρ (x - m) + Sqrt[(x - m) ^2 + σ^2]);
```

In[•]:=
```
Simplify[V'[x]]
```

Out[•]=

$$b\left(\rho+\frac{-m+x}{\sqrt{(m-x)^2+\sigma^2}}\right)$$

In[•]:=
```
Simplify[V''[x]]
```

Out[•]=

$$\frac{b\,\sigma^2}{\left(m^2-2\,m\,x+x^2+\sigma^2\right)^{3/2}}$$